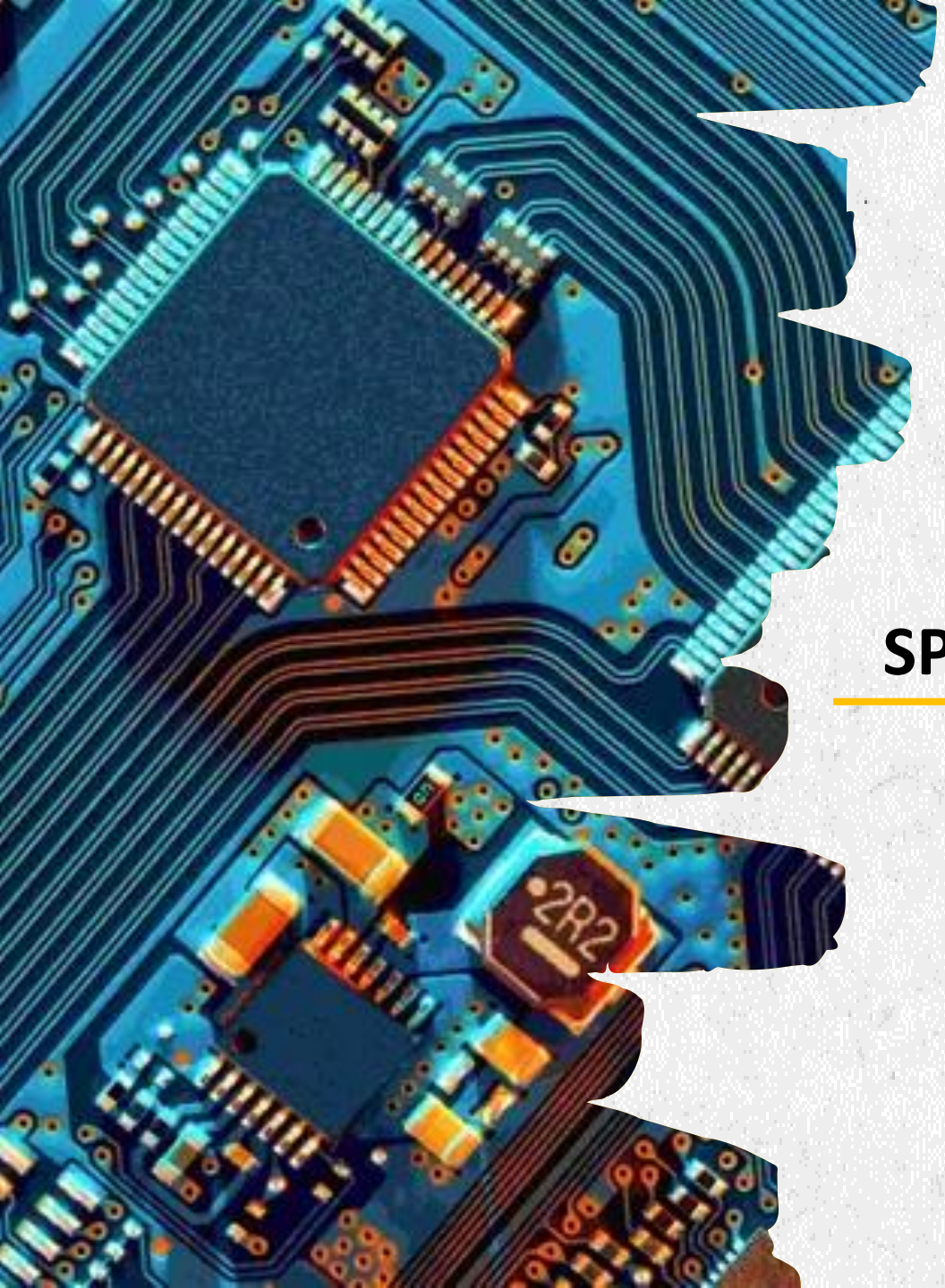


CHAPTER 5

SPECIAL-PURPOSE DIODES

SPECIAL-PURPOSE DIODES

- Last chapter discussed the *general-purpose diodes*.
- However, there are several other types of diodes that are designed for *specific applications*, including

1. Zener diodes 稽納二極體
2. Light-emitting diodes 發光二極體
3. Photodiodes 光電二極體

Discussed in this Chapter

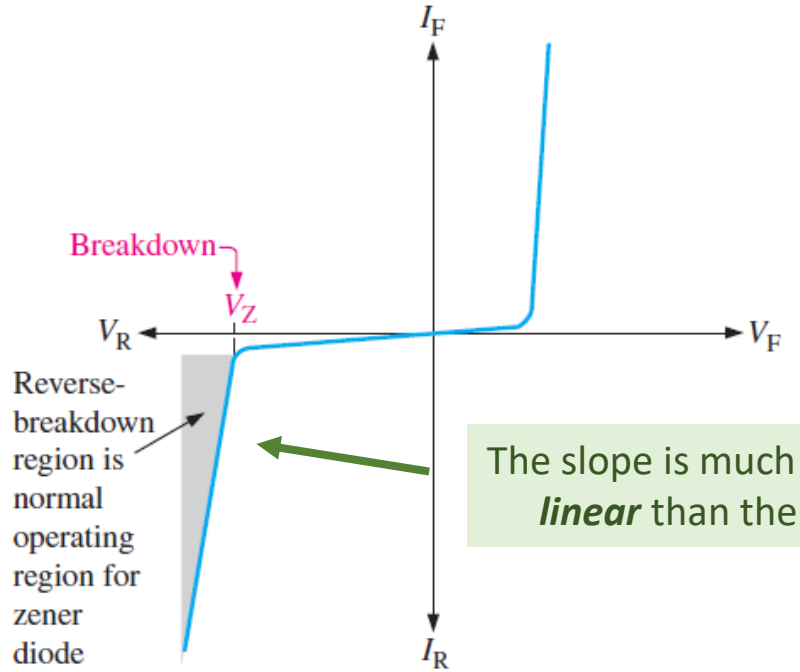
4. Varactor diodes 變容二極體
5. Laser diodes 雷射二極體
6. Schottky diodes 蕭特基二極體
7. Tunnel diodes 隧道二極體
8. Current regulation diodes 穩流二極體

Read them yourself, if interested

ZENER DIODES

ZENER DIODES

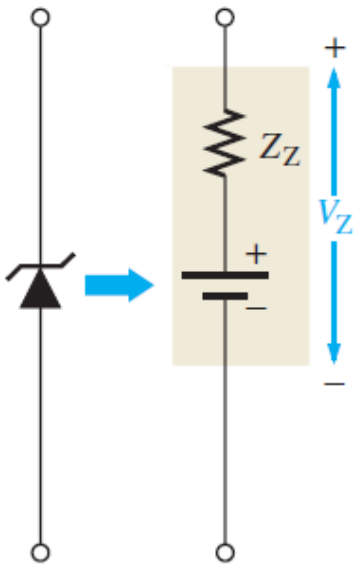
- A *zener diode* 稽納二極體 is a silicon *pn* junction device that is designed for operation in the *reverse-breakdown region* 逆向崩潰範圍. The *breakdown voltage* 崩潰電壓 is set by *controlling the doping level* during manufacture.
- *Zener diodes* are commercially available with *breakdown voltages from less than 1 V to more than 250 V* with specified tolerances 公差 from 1% to 20%.

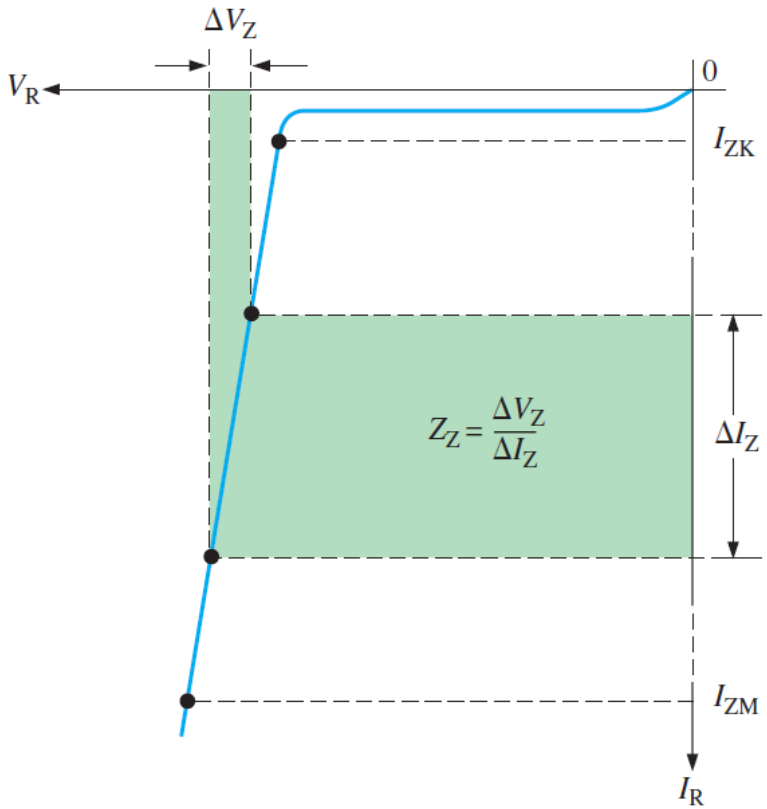


The slope is much ***more gradual and linear*** than the general diodes.

ZENER EQUIVALENT CIRCUITS

- At the *reverse breakdown region* 逆向崩潰範圍, zener diodes can be approximated by a *constant voltage drop* (like *potential barrier*) with a *zener impedance* 稽納阻抗 Z_z, producing the zener voltage 稽納電壓 V_z.





FAIRCHILD SEMICONDUCTOR®

1N4728A - 1N4764A

Zeners

DO-41 Glass case
COLOR BAND DENOTES CATHODE

Absolute Maximum Ratings * T_a = 25°C unless otherwise noted

Symbol	Parameter	Value	Units
P_D	Power Dissipation @ TL ≤ 50°C, Lead Length = 3/8"	1.0	W
	Derate above 50°C	6.67	mW/°C
T_J, T_{STG}	Operating and Storage Temperature Range	-65 to +200	°C

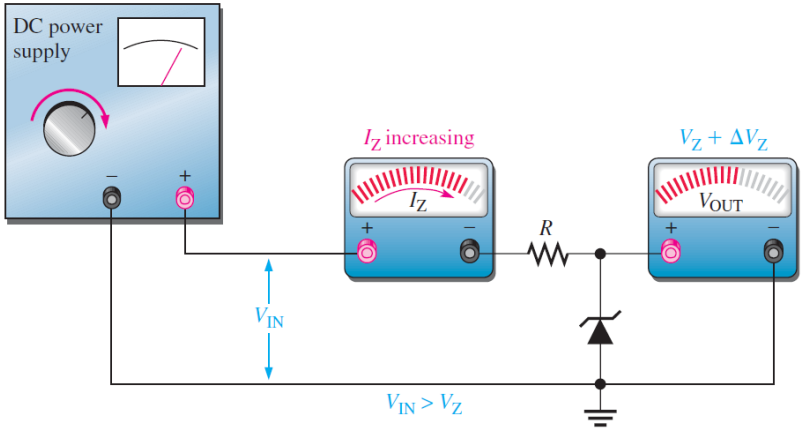
* These ratings are limiting values above which the serviceability of the diode may be impaired.

Electrical Characteristics T_a = 25°C unless otherwise noted

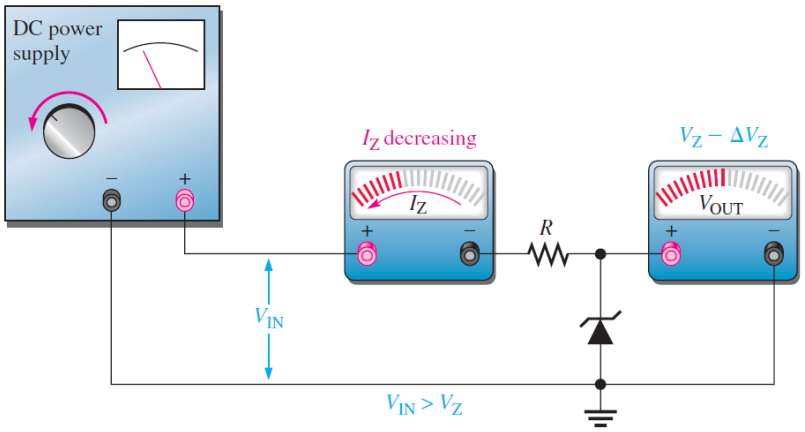
Device	V_Z (V) @ I_Z (Note 1)			Test Current I_Z (mA)	Max. Zener Impedance			Leakage Current	
	Min.	Typ.	Max.		Z_Z @ I_Z (Ω)	Z_{ZK} @ I_{ZK} (Ω)	I_{ZK} (mA)	I_R (μA)	V_R (V)
1N4728A	3.315	3.3	3.465	76	10	400	1	100	1
1N4729A	3.42	3.6	3.78	69	10	400	1	100	1

ZENER DIODE APPLICATIONS

- *Zener diodes* are used in regulating a constant DC level at the output 穩壓, while the *input voltage* or the *output load* are *variable* 變化. This is called *zener regulation*.



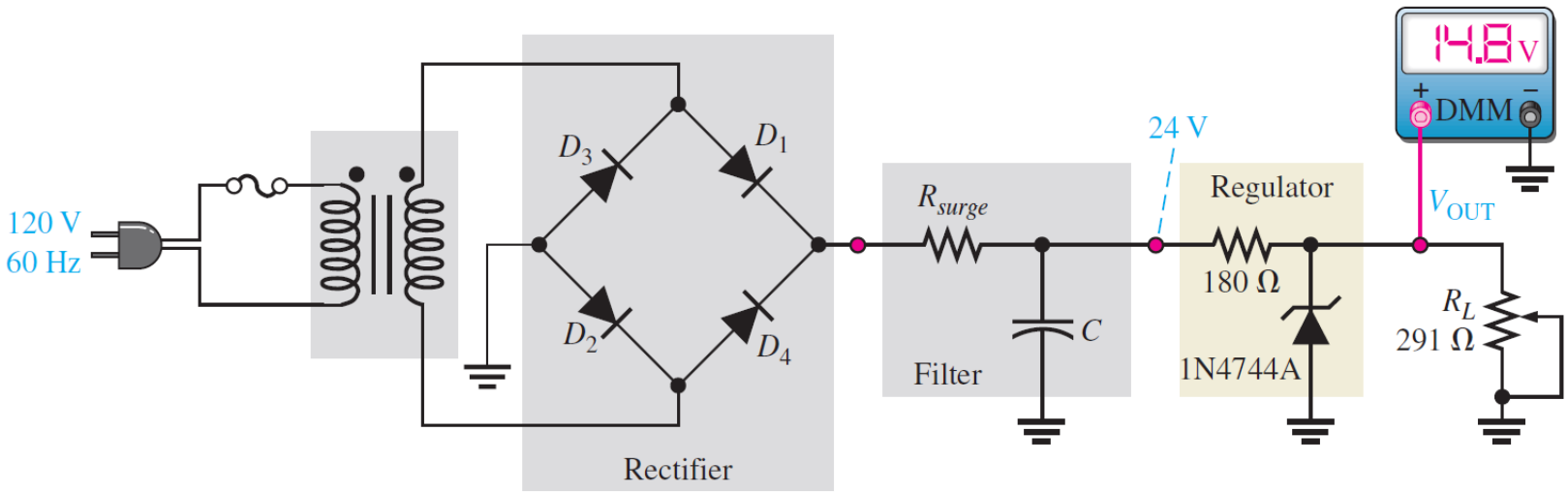
As the input voltage increases, the output voltage remains nearly constant.



As the input voltage decreases, the output voltage remains nearly constant.

- If the zener impedance 稽納阻抗 Z_z is ideally small, the zener diode can perfectly maintain the constant DC level at the output. However, they are limited to applications that *require only low current to the load*.

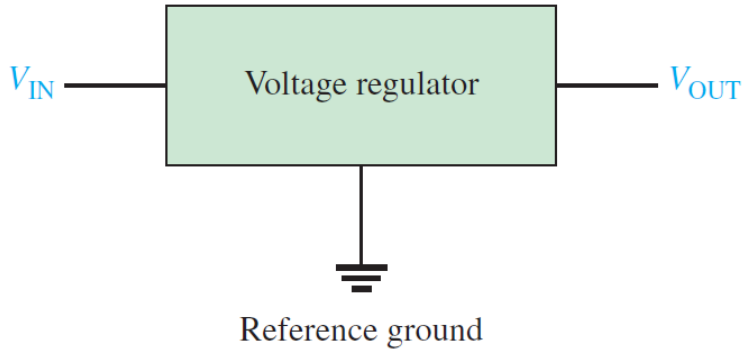
ZENER DIODE APPLICATIONS

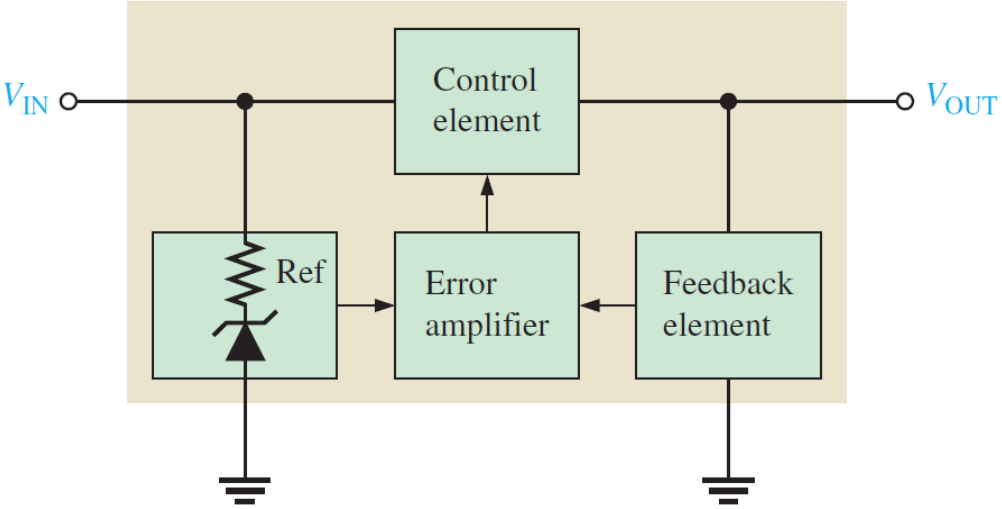


- The above circuit is the DC converter for +15 VDC supply, without a chip 78XX.
- Due to the use of the Zener diode, instead of IC 78XX:
  1. The price is cheaper.
  2. The circuit CANNOT supply high current to the load.

ZENER DIODES IN VOLTAGE REGULATOR

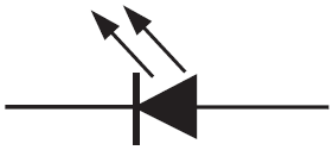
- To achieve better regulation and provide for greater variations in load current, the ***zener diode*** is combined as a key element with other circuit components to create ***a 3-terminal linear voltage regulator 穩壓器***.
- The ***zener*** to provide ***a reference voltage 參考電壓 for an internal amplifier***. The ***DC output voltage*** is always ***less than the input voltage*** ($V_{IN} > V_{OUT}$).





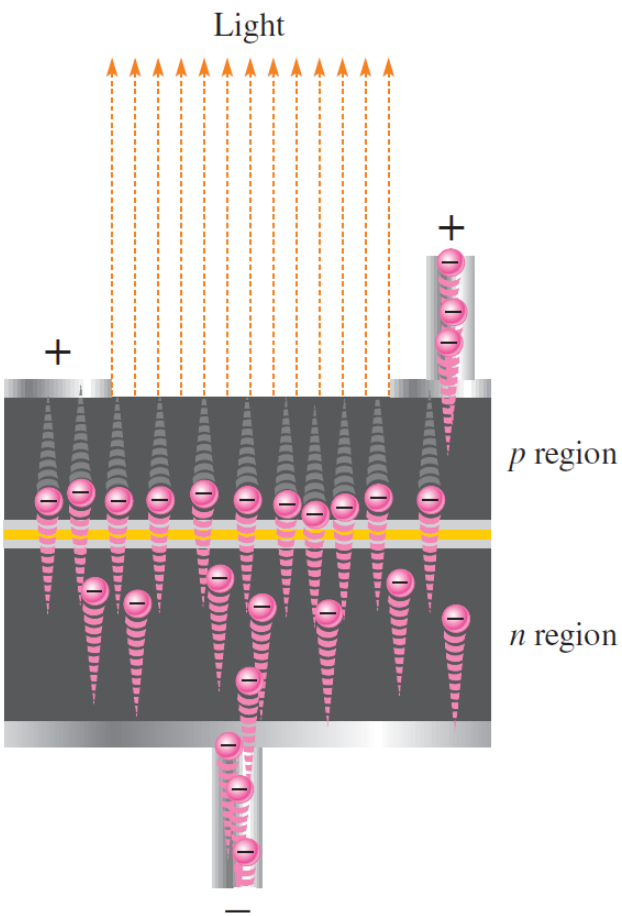
OPTICAL DIODES AND SENSORS

LIGHT-EMITTING DIODE (LED)



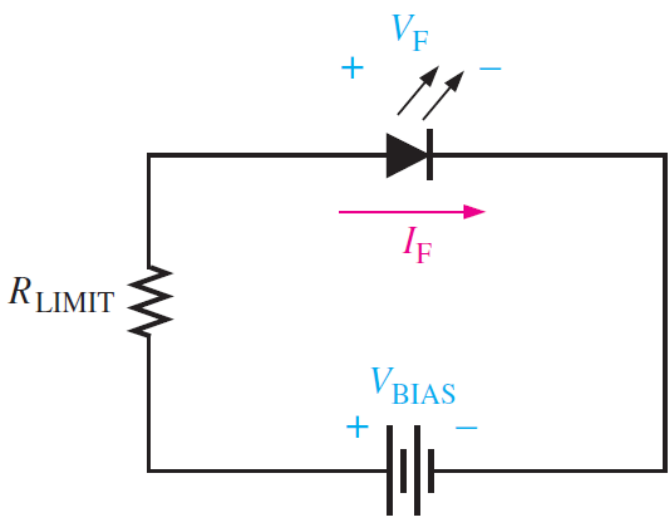
- An ***LED (Light Emitting Diode、發光二極體)*** *emits light* when *electrons 電子 and holes 電子孔 recombine* in its semiconductor material, releasing *energy in the form of photons 光子* through ***electroluminescence 電致發光***.

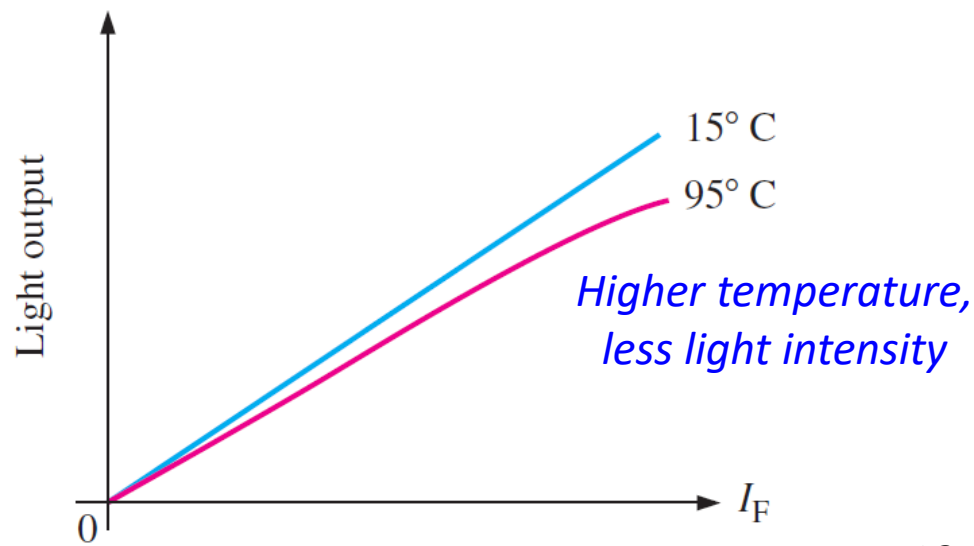
- ***Various impurities*** are added during the ***doping process 摻雜*** to establish the ***wavelength 波長 (color)*** of the emitted light.



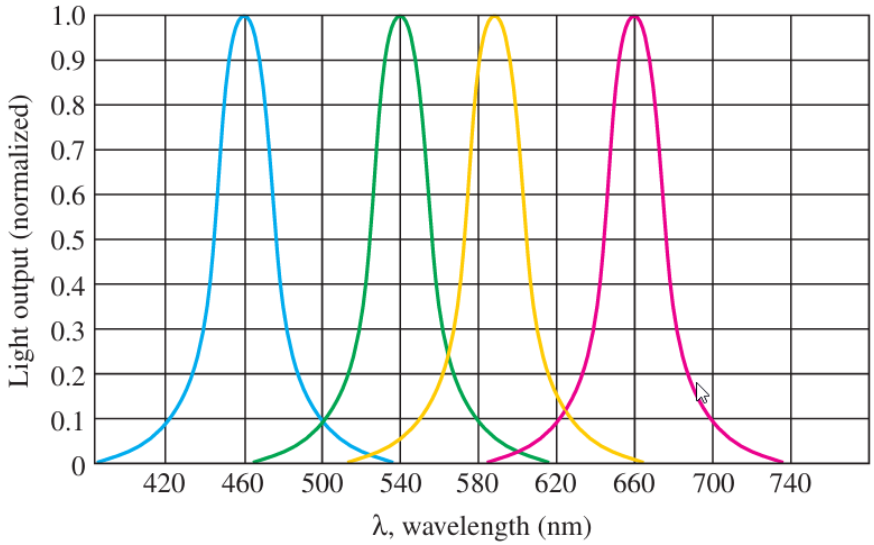
LED BIASING

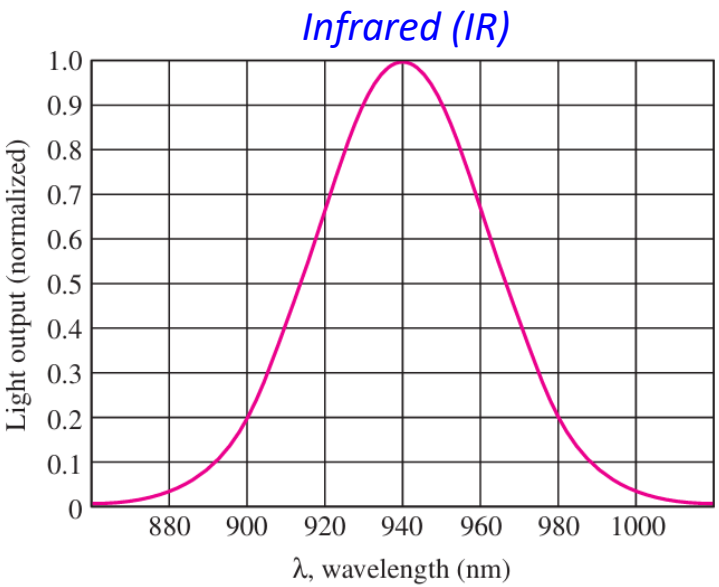
- The forward voltage 順向偏壓 across an LED is greater than for a silicon diode. Typically, the maximum V_F for LEDs is between 1.2 V and 3.2 V, depending on the material.
- ***Reverse breakdown* 逆向崩潰電壓** for an LED (typically, 3 V to 10 V) is much less than for a silicon diode.
- The LED emits light in response to a sufficient forward current. The *light intensity* 光強度 *(power output)* is directly proportional to the *forward current*.

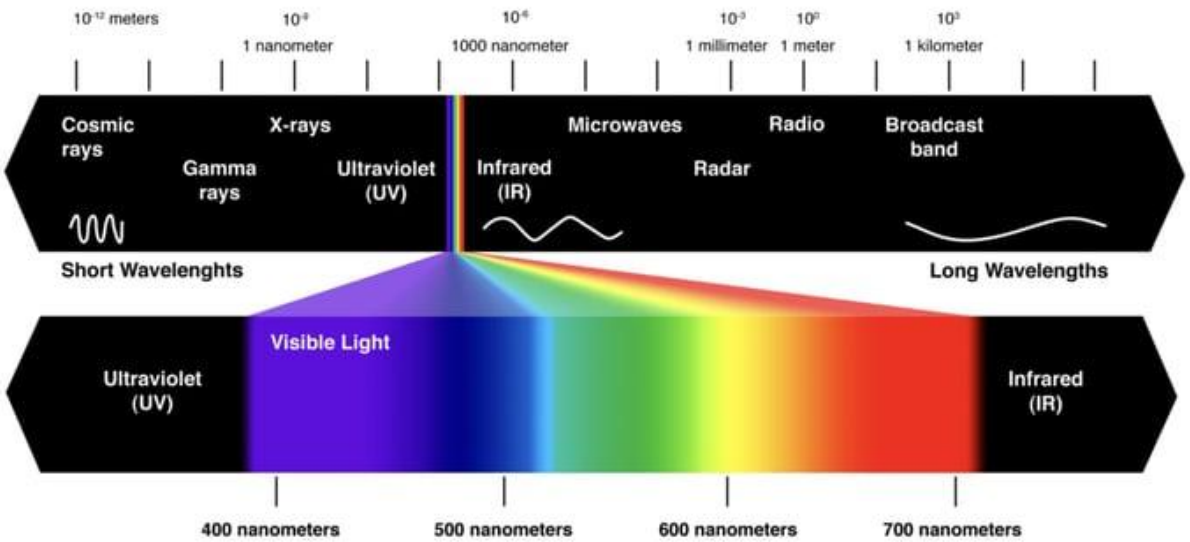




- An LED emits light over a specified range of wavelengths as indicated by the spectral output curves.



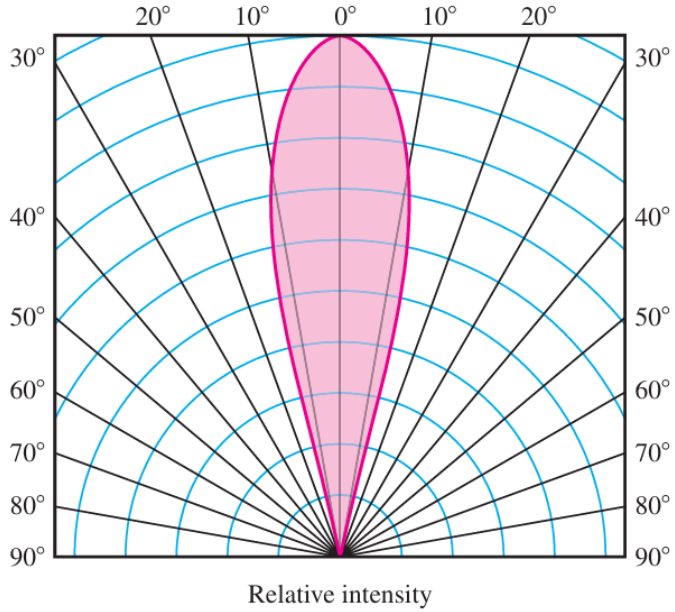




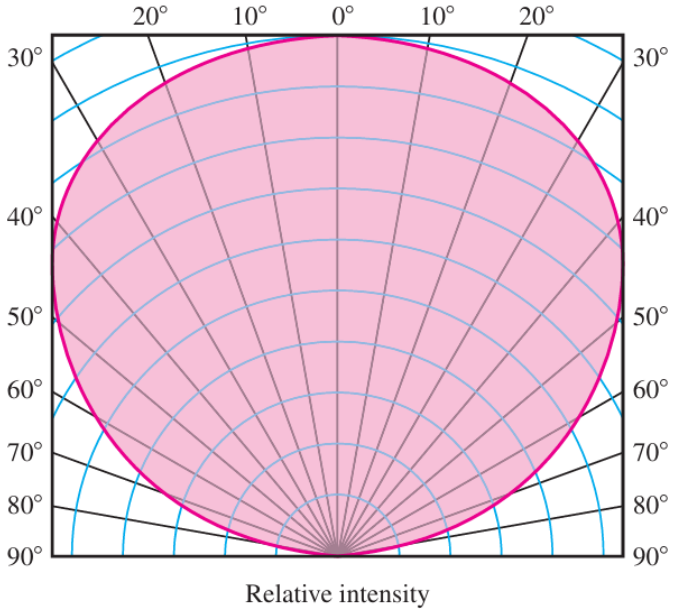
RADIATION PATTERNS

- LEDs are directional light sources 方向光, which have specific radiation patterns. It can be altered by the shape of the surface and by lenses 鏡片 and diffusion films 擴散膜 to favor a specific direction.

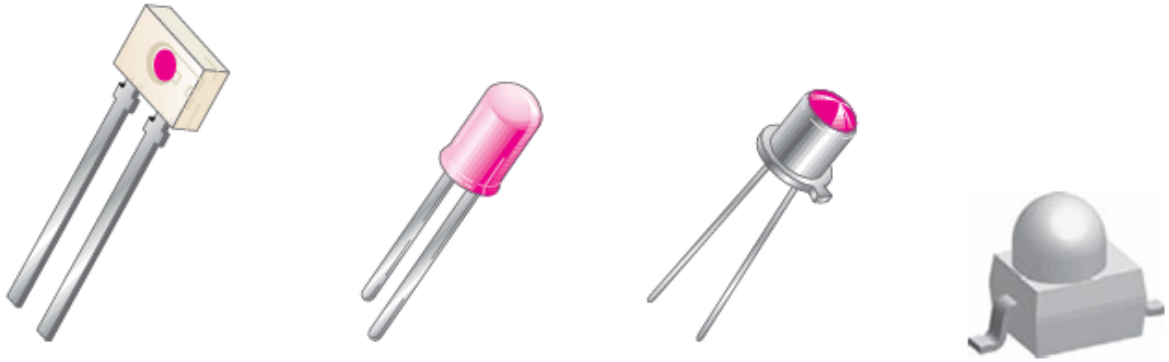
Narrow viewing angle



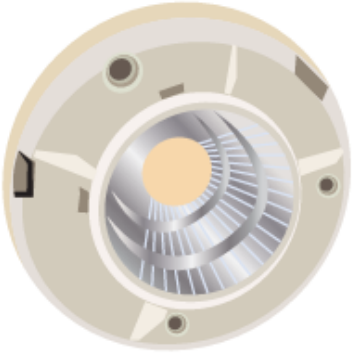
Wide viewing angle



LED LAMPS



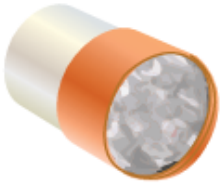
(a) Typical small LEDs for indicators



Helion 12 V overhead light with socket and module

120 V, 3.5 W screw base for low-level illumination

120 V, 1 W small screw base candelabra style



6 V, bayonet base for flashlights, etc.

(b) Typical LEDs for lighting applications

Absolute Maximum Ratings

T_{amb} = 25°C, unless otherwise specified

Parameter	Test condition	Symbol	Value	Unit
Reverse Voltage		V_R	5	V
Forward current		I_F	100	mA
Peak Forward Current	t_p/T = 0.5, t_p = 100 µs	I_{FM}	200	mA
Surge Forward Current	t_p = 100 µs	I_{FSM}	0.8	A
Power Dissipation		P_V	190	mW
Junction Temperature		T_j	100	°C
Operating Temperature Range		T_{amb}	- 40 to + 85	°C

Basic Characteristics

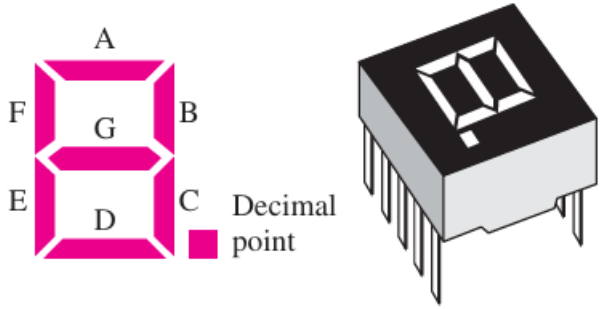
T_{amb} = 25°C, unless otherwise specified

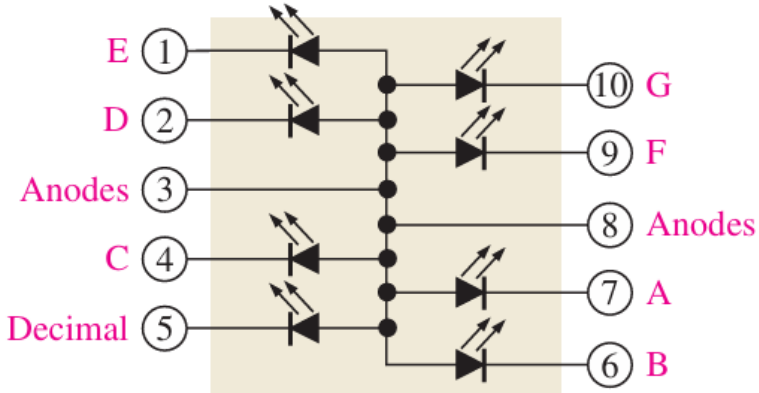
T_{amb} = 25°C, unless otherwise specified

Parameter	Test condition	Symbol	Min	Typ.	Max	Unit
Forward Voltage	I_F = 20 mA	V_F		1.3	1.5	V
	I_F = 1 A, t_p = 100 µs	V_F		2.4		V
Temp. Coefficient of V_F	I_F = 1.0 mA	TK_{VF}		- 1.7		mV/K
Reverse Current	V_R = 5 V	I_R			10	µA
Junction capacitance	V_R = 0 V, f = 1 MHz, E = 0	C_j		160		pF
Radiant Intensity	I_F = 20 mA	I_e	2.5	5	13	mW/sr
	I_F = 100 mA, t_p = 100 µs	I_e		25		mW/sr
Radiant Power	I_F = 100 mA, t_p = 20 ms	ϕ_e		35		mW
Temp. Coefficient of·ϕ_e	I_F = 20 mA	$TK\phi_e$		- 0.6		%/K
Angle of Half Intensity		ϕ		± 17		deg
Peak Wavelength	I_F = 20 mA	λ_p		870		nm
Spectral Bandwidth	I_F = 20 mA	$\Delta\lambda$		40		nm
Temp. Coefficient of λ_p	I_F = 20 mA	$TK\lambda_p$		0.2		nm/K
Rise Time	I_F = 20 mA	t_r		30		ns
Fall Time	I_F = 20 mA	t_f		30		ns

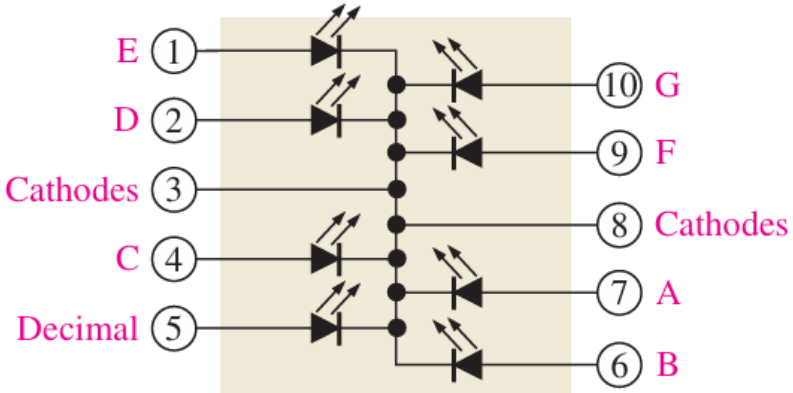
LED APPLICATIONS

- A common type of display device using LEDs is *the 7-segment display 七段顯示器*.
- Combinations of the segments form the *ten decimal digits 十進制*. Each segment in the display is an LED.





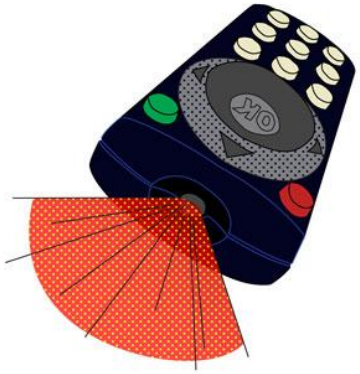
(b) Common anode



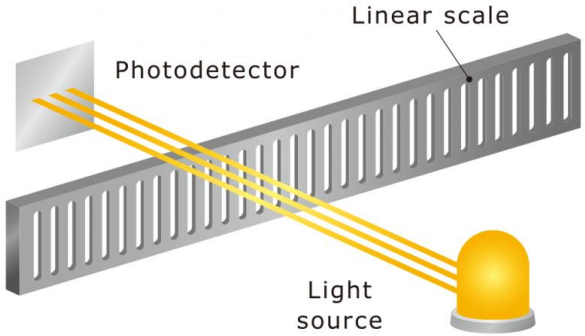
(c) Common cathode

LED APPLICATIONS

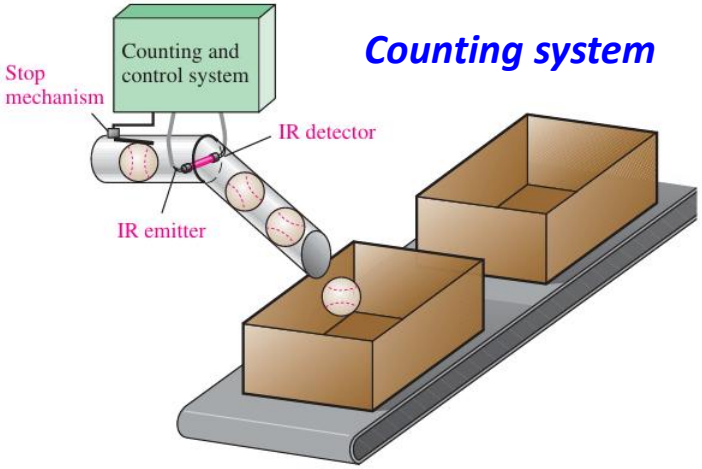
- *Infrared (IR、紅外線) LEDs* are also used in a wide range of applications, although the *IR light is invisible 隱形*. *IR LEDs* are usually works with *photodiodes 光電二極體* as *signal transmitter/receiver 訊號傳送/接收器*.



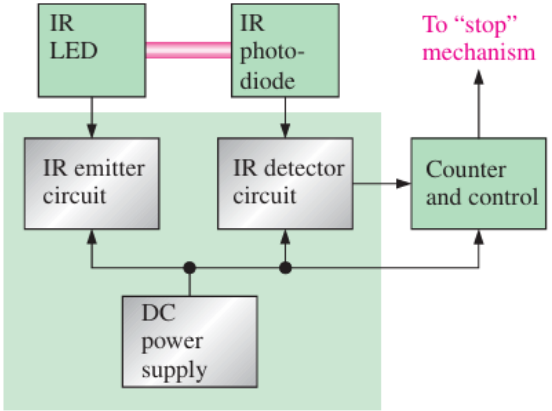
Remote controller
(TVs, DVD player, gate opener)



Position measurement



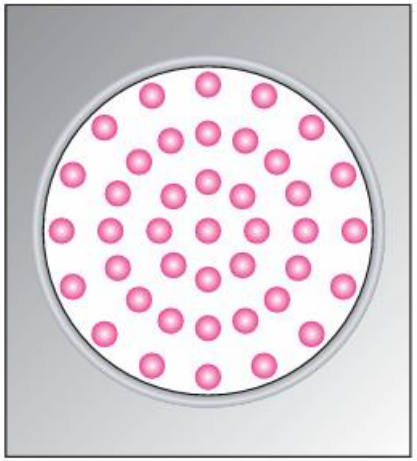
Counting system

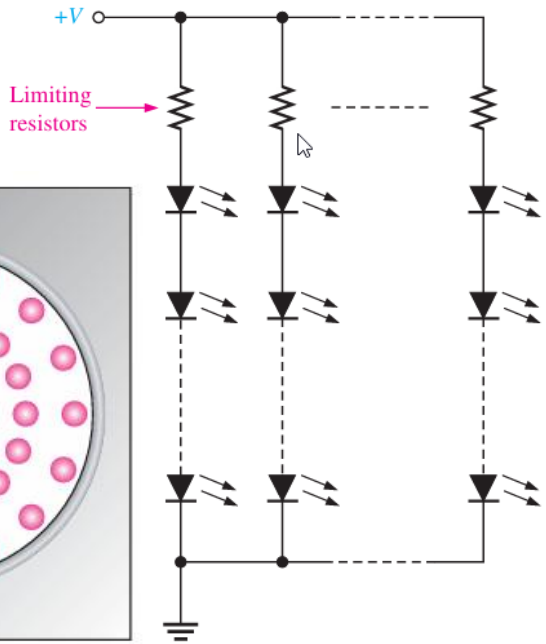


TRAFFIC LIGHTS

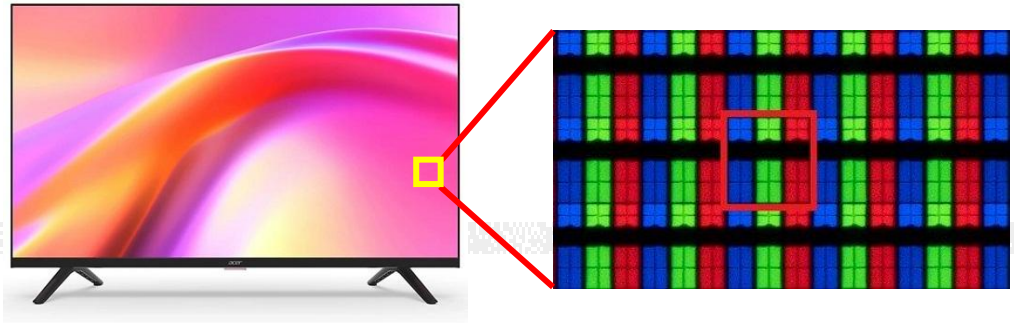
- *Arrays 矩陣 of tiny LEDs* form the red, yellow, and green lights in *a traffic light unit*.
- *An LED array 矩陣*LED has three major advantages over the *incandescent bulb 白熾燈*:
  - Brighter light
  - Longer lifetime 壽命 (years vs months)
  - Less energy consumption 能耗 (about 90% less)

Incandescent bulb
（白熾燈）

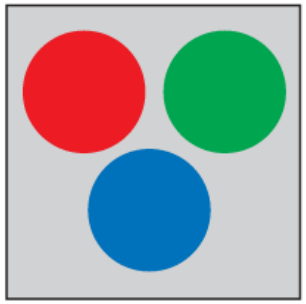


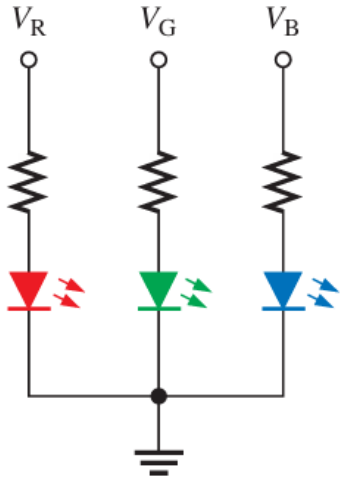


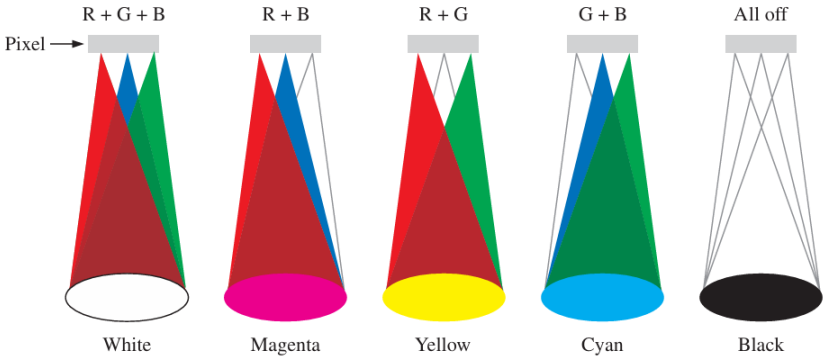
LED DISPLAYS



- *LEDs* are widely used in large and small *signs and message boards* for both indoor and outdoor uses, including *TVs*.
- Signs can be single-color, multicolor, or full-color. *Full-color screens* use a tiny grouping of high-intensity *red, green, and blue (RGB) LEDs* to form *a pixel 畫素*.

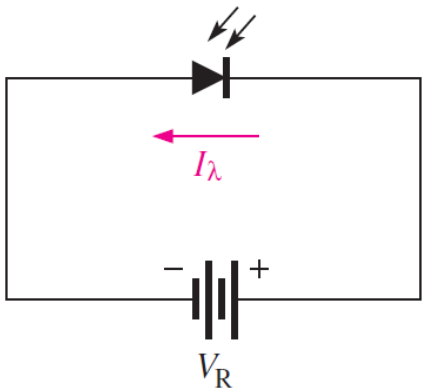


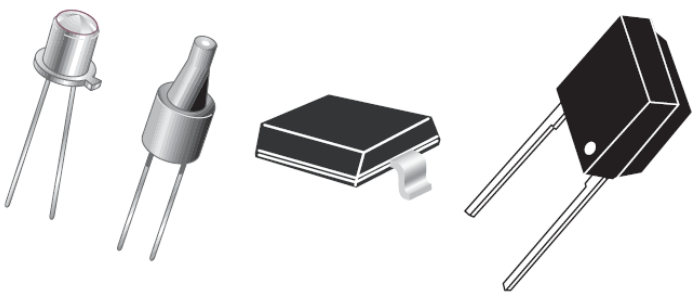


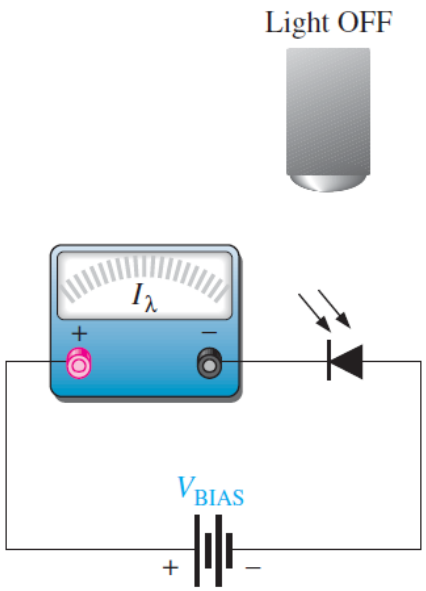


PHOTODIODES

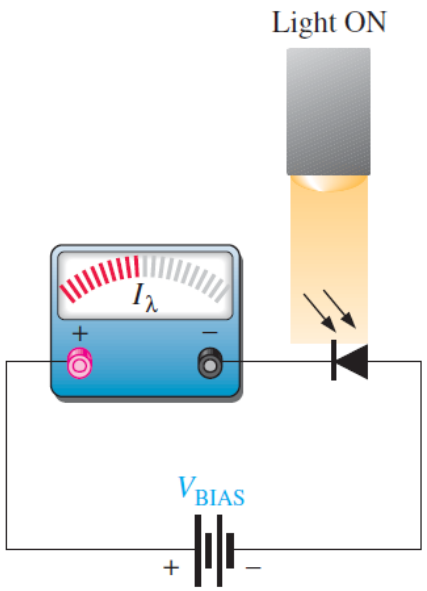
- The *photodiode 光電二極體* is a device that operates in *reverse bias*. The photodiode has *a small transparent window that allows light to strike the pn junction*.
- The *pn* junction is exposed to light, the *reverse current 逆向電流 increases with the light intensity 光強度* .







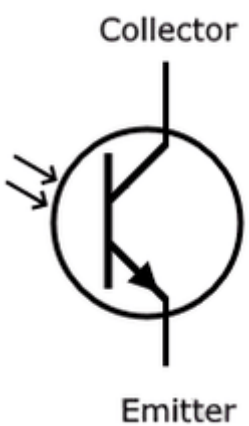
(a) No light, no current except negligible dark current

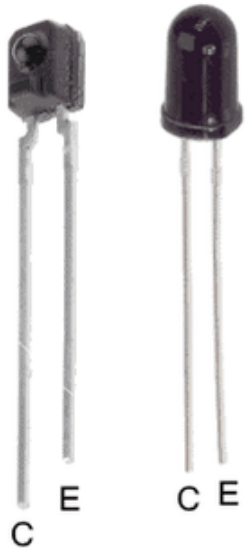


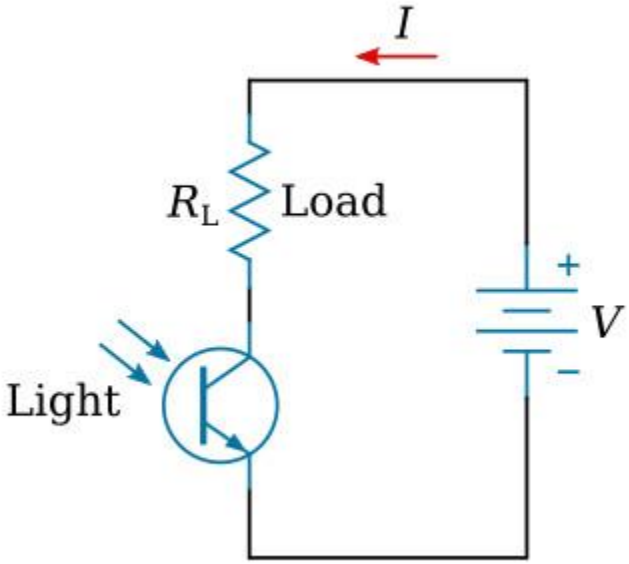
(b) Where there is incident light, resistance decreases and there is reverse current.

PHOTOTRANSISTORS

- A phototransistor 光電晶體 is a light-sensitive, three-terminal active semiconductor device that functions as a transistor 電晶體 where the base current is provided by incident light.
- It acts as a light-controlled switch, offering higher sensitivity than photodiodes.

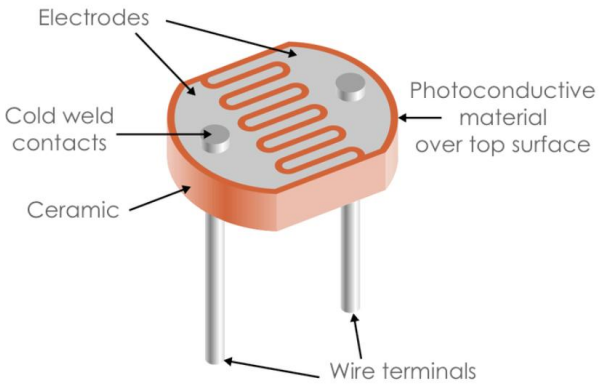


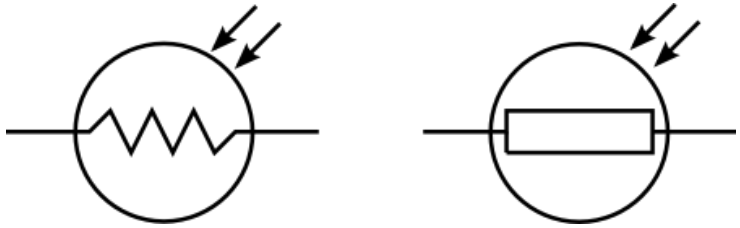


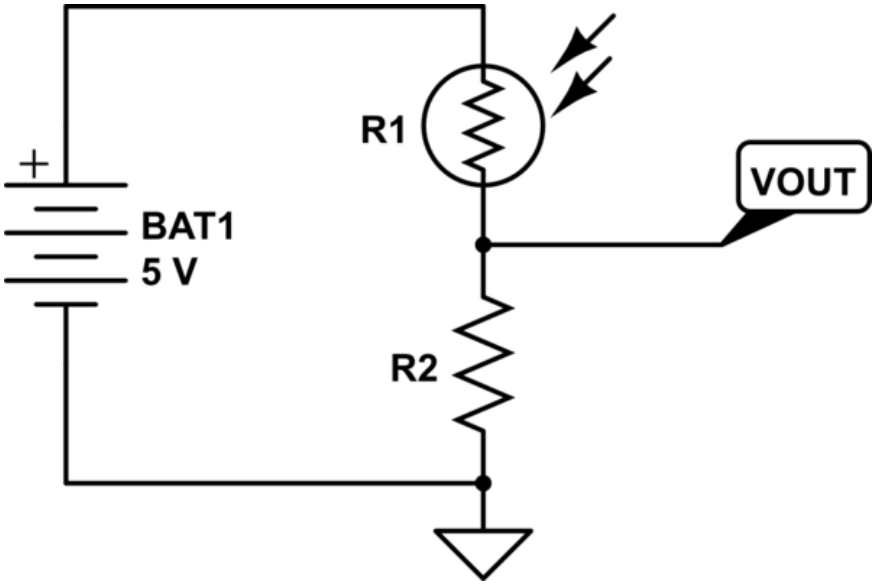


PHOTORESISTOR

- A photoresistor 光敏電阻, or light dependent resistor (LDR), is a passive electronic component whose resistance decreases as light intensity on its surface increases, and increases in darkness.

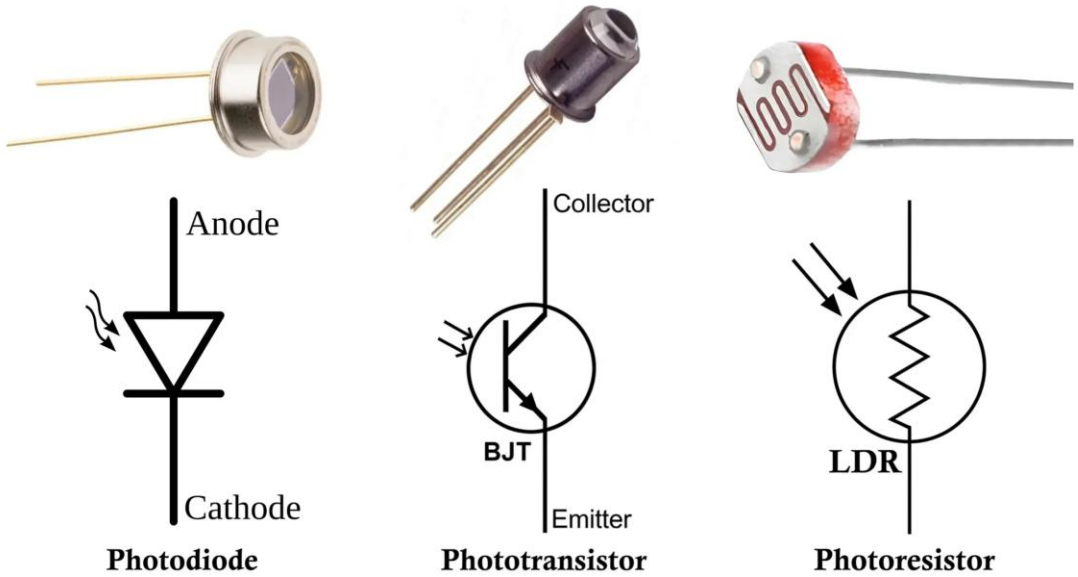






COMPARISON

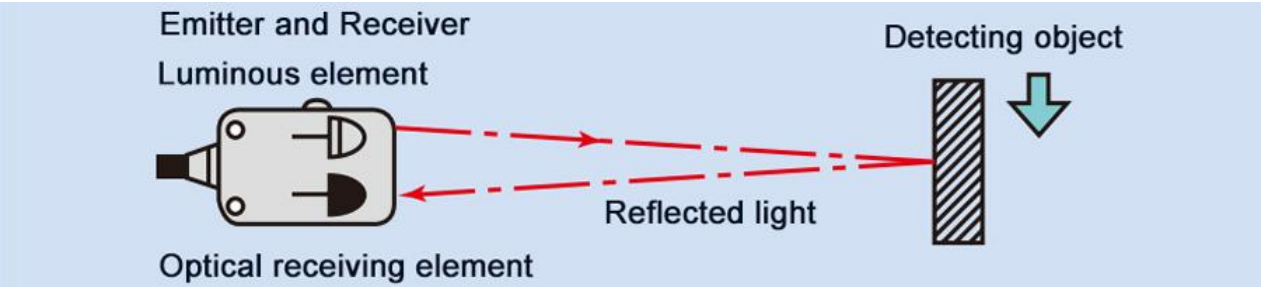
- Photodiode – current flow depending upon the light intensity.
- Phototransistor – acts as switch on/off according to the exposed light.
- Photoresistor (or LDR) – resistance inversely proportional to the light intensity.



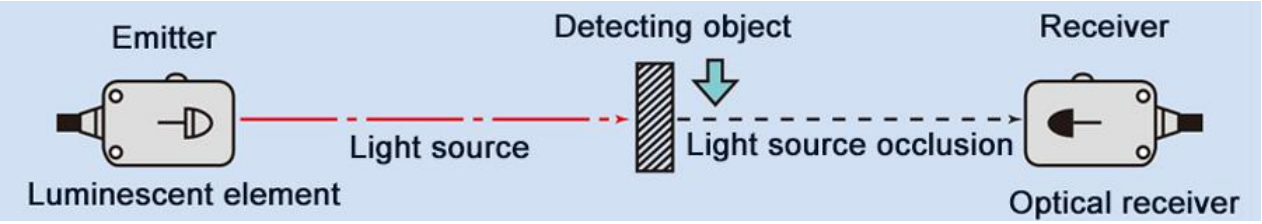
APPLICATIONS

- A pair of infrared (IR) LED and phototransistor is often used in various sensor modules.

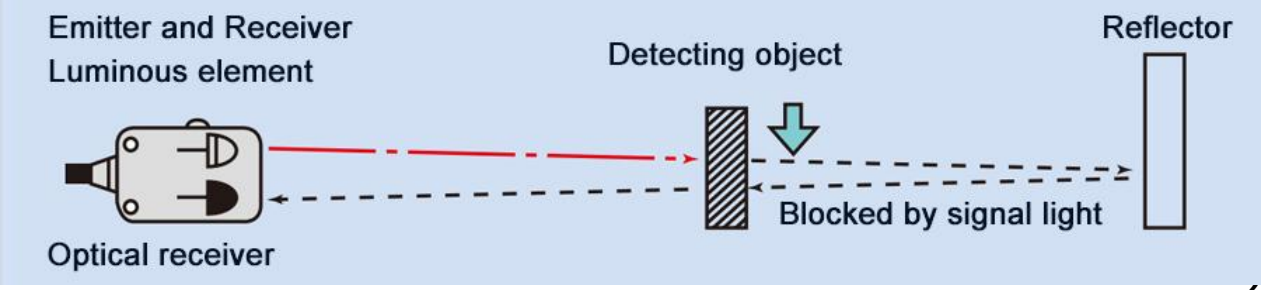
Diffuse reflection



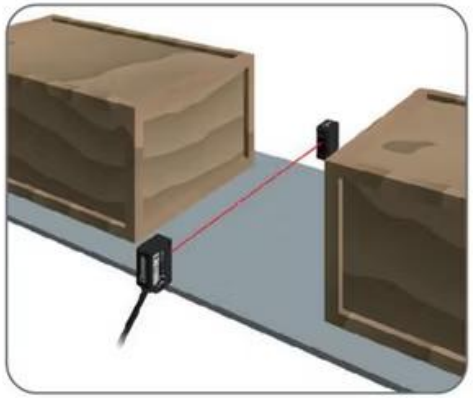
Through beam



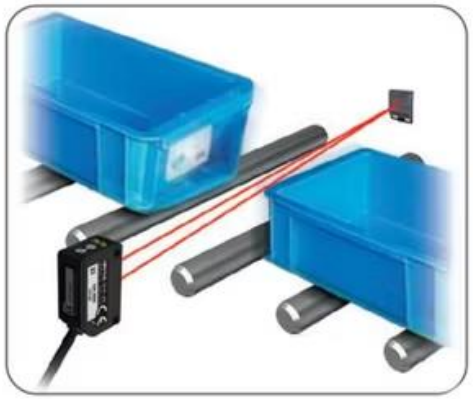
Retro reflection



APPLICATIONS



PF21T

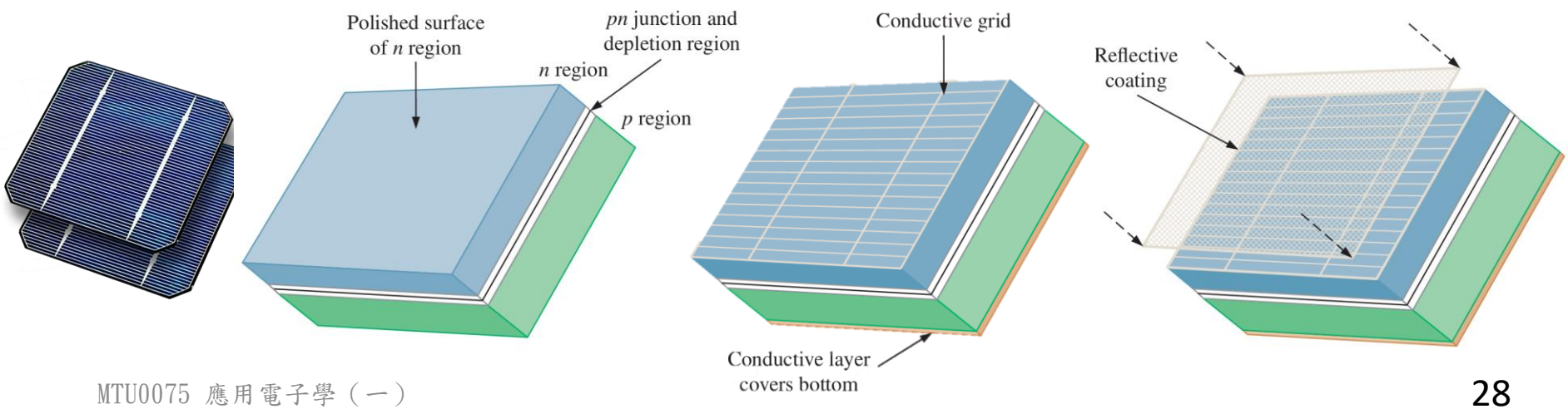


PF21R

SOLAR CELLS

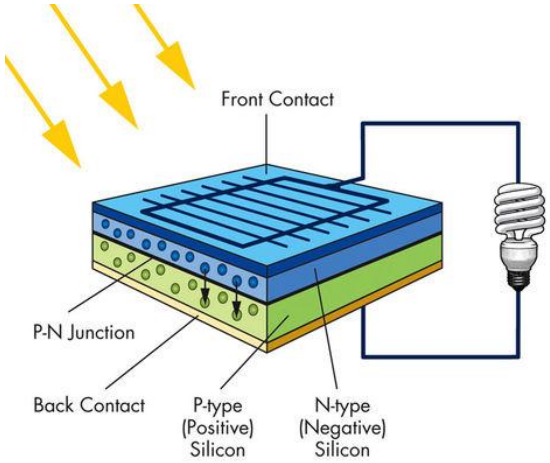
SOLAR CELLS

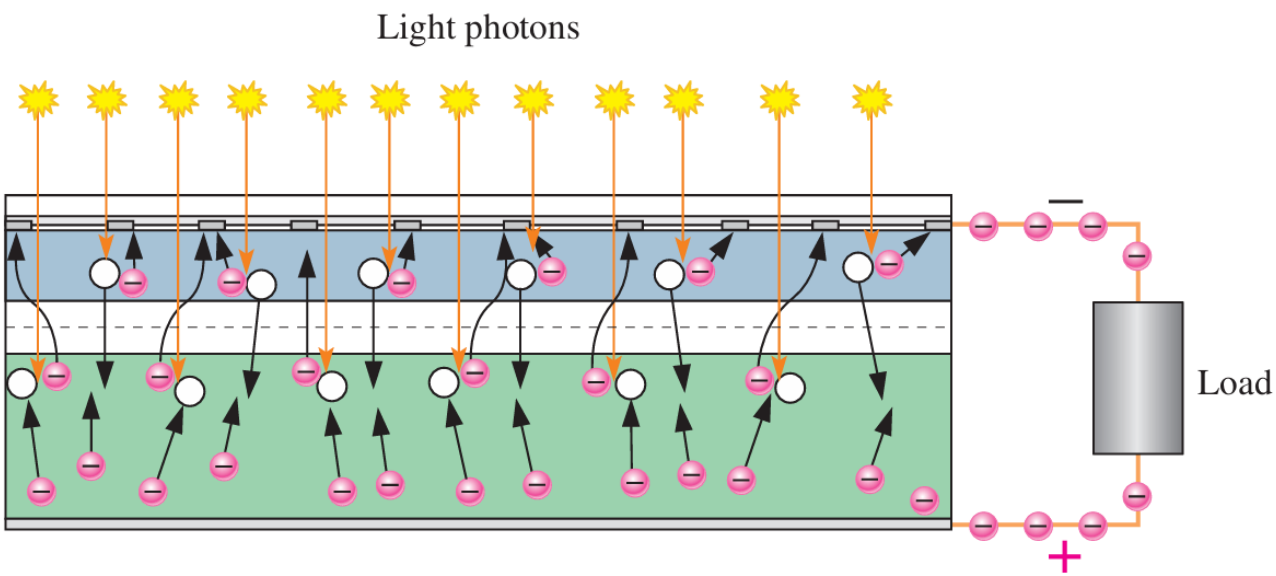
- *Solar cell* 太陽能電池 consists of a thin layer of silicon that has been doped to create a *pn* junction. The silicon wafer is doped so that the *n* region is much thinner than the *p* region to permit light penetration.
- *Conductive grid* 導電網 is to conduct the generated electricity to an external load.
- *Anti-reflective coating* 抗反射膜 allows the solar cell to absorb as much of the sun's energy as possible by reducing the amount of light energy reflected away from the solar cell.



SOLAR CELLS

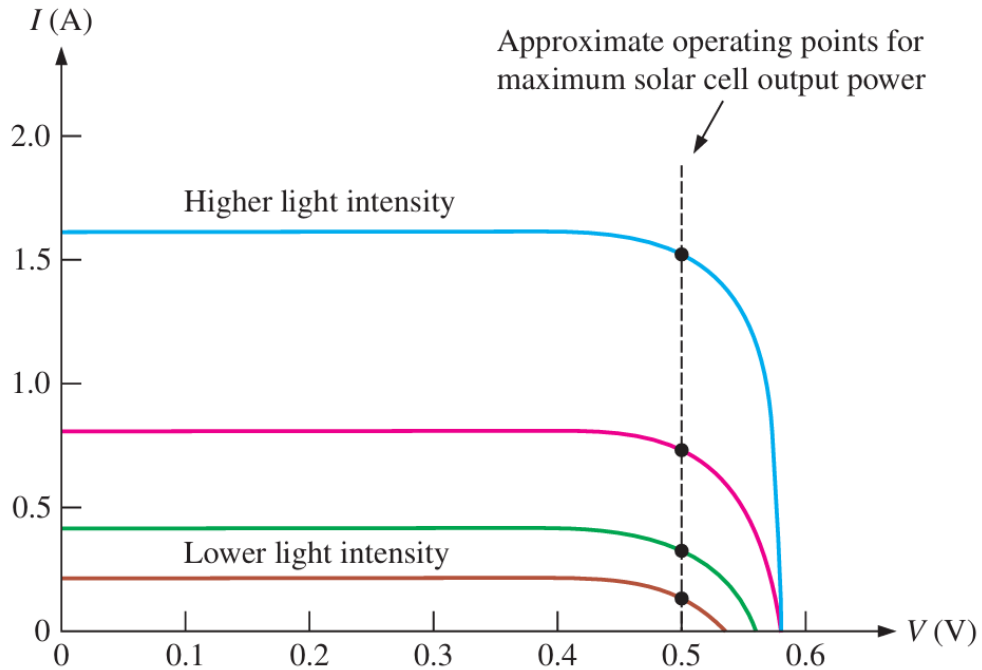
- The ***photovoltaic effect 光伏效應*** is the basic physical process by which a solar cell converts *sunlight into electricity*.
- Sunlight contains ***photons 光子*** that are sufficient to create electron-hole pairs in the *n* and *p* regions. Electrons accumulate in the *n* region and holes accumulate in the *p* region, producing a potential difference (voltage) across the cell.





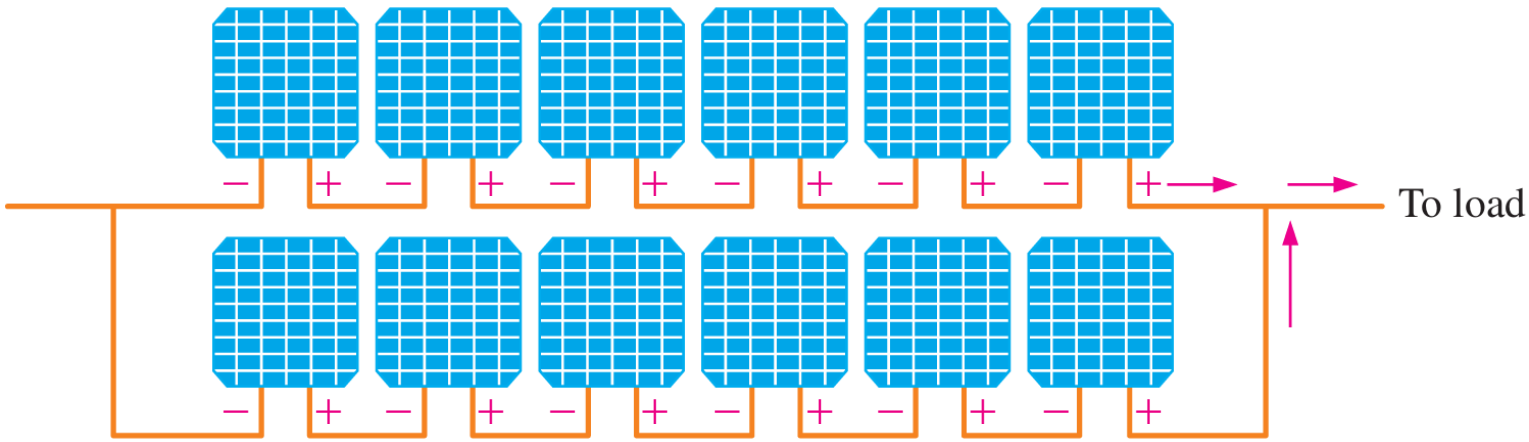
SOLAR CELLS

- Solar cells太陽能電池 are typically 100 cm^2 to 225 cm^2 in size.
- The usable voltage from solar cells is approximately 0.5 V to 0.6 V.
- Terminal voltage is only slightly dependent on the intensity of light radiation, but the current *increases* with light intensity.

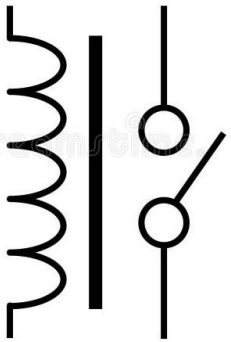
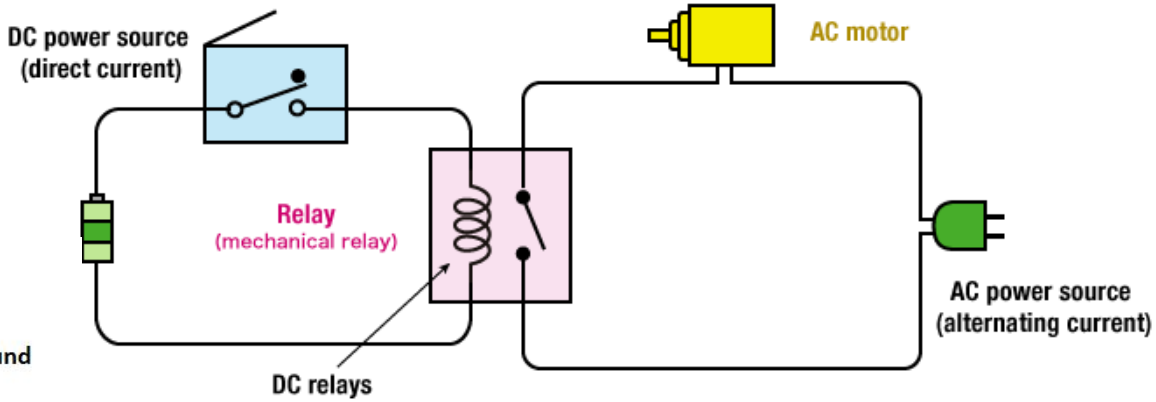


SOLAR CELLS

- A single solar cell is impractical for most applications because it can produce only about 0.5 V to 0.6 V.
- To produce *higher voltages*, multiple solar cells are connected in series.
- To produce *higher current capacity*, series cells are connected in parallel.



RELAY

RELAY

- A relay (繼電器) is an electrically operated switch that uses a low-power signal to control a high-power circuit, offering electrical isolation between them.
- These components, which include electromechanical and solid-state types, enable microcontrollers to manage heavy loads like motors and lights.